ZCAS University

CIT 3241 BUSINESS SYSTEMS ANALYSIS

MID TERM TEST

20TH APRIL 2026

10:30 to 12:00

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Answer any **TWO** questions from three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

QUESTION ONE

The IT industry continues to experience high failure rates in information system projects. Studies show that a significant number of large IT projects are cancelled before completion, many exceed their budgets substantially, and most completed systems deliver only a fraction of their intended functionality. In addition, several information systems are under-utilized because they fail to meet user needs, while legacy systems increasingly hinder organizational growth and adaptability.

(a) Using the information above, identify and explain seven (7) key reasons why systems analysis is essential in the development of information systems. **[7 marks]**

(b) Discuss eight (8) ways in which effective systems analysis can help organizations reduce project failure, control costs, improve user satisfaction, and manage legacy systems. **[8 marks]**

c) Agile is a group of software development methodologies such as Scrum, Extreme Programming (XP), and Lean, which emphasize small incremental development, adaptability to change, and close collaboration among stakeholders.

(i) Define agility in the context of software development. **[2 marks]**

(ii) Agile methodologies are guided by four core values. Explain any four (4) of the following Agile values, showing how each contributes to successful software development:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan **[8 marks]**

QUESTION TWO

Extreme Programming (XP) is an agile software development methodology created by Kent Beck. It focuses on delivering high-quality software while responding quickly to changing customer requirements through continuous planning, close customer involvement, and short development cycles.

(a) Define Extreme Programming (XP) and explain XP Planning, highlighting the two key questions it addresses in software development and how XP emphasizes project steering over exact prediction. **[8 marks]**

(b) Describe the two key planning practices in XP, namely Release Planning and Iteration Planning. In your answer, explain the role of the customer and programmers, the importance of short iterations, and how these practices improve visibility, control, and value delivery in XP projects. **[7 marks]**

(c) Explain the importance of requirements determination in ensuring project success. **[5 marks]**

(d) Classify system requirements into Functional and Non-Functional requirements, giving one example of each. **[5 marks]**

QUESTION THREE

Requirement's elicitation has evolved significantly over time due to changes in technology, business environments, and development methodologies, leading to a paradigm shift from traditional to modern approaches.

(a) Explain the key characteristics of traditional requirements elicitation approaches. **[4 marks]**

(b) Describe the main features of **modern requirements elicitation** approaches and how they differ from traditional methods. **[4 marks]**

(c) Discuss the **reasons for the paradigm shift** from traditional to modern requirements elicitation and explain how this shift improves project success. **[7 marks]**

(d) Describe requirements elicitation techniques used by systems analysts. In your answer, explain any **three (3)** of the following techniques and state when each is most appropriate to use:

- Interviews (structured, semi-structured, or unstructured)
- Questionnaires
- Focus groups
- Observation **[10 marks]**

THE END